

REDA EL MIKKAWI
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Saida, Lebanon

EDUCATION

Lebanese American University (LAU)

BE in Mechanical Engineering

Byblos, Lebanon

Expected graduation date June 2026

Houssam Hariri High School

Lebanese Baccalaureate, General Science Section

Saida, Lebanon

2021

Experience

Internship Trainee – Automotive Maintenance

December 22, 2025 – January 13, 2026

Porsche Lebanon, Lebanon

- Assisting in vehicle diagnostics, mechanical maintenance, and service operations.
- Working on brake systems, suspension, engine components, and OBD analysis.
- Gaining practical experience in the workflow of modern automotive workshops.

Internship Trainee – Automotive Maintenance

July1, 2025 – August28, 2025

RYMCO (Rasamny Younis Motor Company), Lebanon

- Assisted in vehicle diagnostics, mechanical maintenance, and service operations.
- Worked on brake systems, suspension, engine components, and OBD analysis.
- Gained practical experience in the workflow of modern automotive workshops.

Internship Trainee – HVAC & MEP Design

June3, 2021 – June30, 2022

Karim ETC (Engineering, Trading & Contracting Company)

- Prepared BOQs and technical drawings for HVAC and MEP systems.
- Used AutoCAD and Revit for system layout and modeling.
- Collaborated with engineers to coordinate mechanical systems across real projects.

Projects

Press Fit Machine-Industrial Automation System (2026)

- Designed and simulated a fully automated servo-pneumatic press-fit assembly system using PLC-based control.
- Integrated 4 double-acting pneumatic cylinders, 8 magnetic reed switches, and 3 photoelectric sensors.
- Developed Sequential Function Chart (SFC) and Ladder Diagram (LAD) logic in WPL software
- Implemented sensor-based transitions for safe, collision-free operation and repeatable insertion of inner cylinder into outer cylinder

Control Systems (2024)

- Designed a self-balancing ball system using control principles. Applied PID control through MATLAB Simulink and Simscape. Integrated ultrasonic sensors and servo motors, and used LabVIEW with FPGA for system accuracy.

Design Automated Bottle Encapsulation Machine (2024)

- Developed a mechanical system for precise bottle capping. Designed the structure using SolidWorks and performed load simulations.

Robotic Hand (2024)

- Created a fully functional robotic hand using sensors and actuators. Focused on mechanical articulation and control logic to simulate human finger motion.

MATLAB Pendulum Analysis (2023)

- Programmed a MATLAB script to analyze and simulate pendulum motion. Extracted displacement, velocity, and acceleration plots for academic research purposes.

Volunteering

Bqosta Volunteers

March 2018 – September 2022

- Organized and managed fundraising events to support the charity's program
- Collaborated with other volunteers to distribute donations
- Helped to increase donations by 20% through targeted outreach efforts

Youth Committee

February 2021 – September 2022

- Enhanced communication and teamwork with diverse volunteers.
- Processed recyclables to reduce waste and promote sustainability.
- Educated the community on recycling practices and waste reduction benefits.

Skills

- **CAD SOFTWARE:** SolidWorks, AutoCAD, Revit.
- **Manufacturing Process:** Machining, Welding, Sheet Metal Fabrication.
- **Programing and Simulation:** MATLAB, Simulink, LabVIEW, Java.
- **Mechanical Design:** Product design, load simulation, control systems integration.
- **Other Skill:** Vehicle diagnostics, HVAC system design, technical drawing, system modeling

Certificates

SolidWorks

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| • Certified SolidWorks Advanced Professional Mold Making (CSWPA-MM) | January 19, 2026 |
| • Certified SolidWorks Advanced Professional Weldment (CSWPA-WD) | July 26, 2025 |
| • Certified SolidWorks Advanced Professional Surfacing (CSWPA-SU) | January 2, 2025 |
| • Certified SolidWorks Associate in Simulation | September 16, 2024 |
| • Certified SolidWorks Advanced Professional Sheet Metal (CSWPA-SM) | August 18, 2024 |
| • Certified SolidWorks Advanced Professional Drawing Tools (CSWPA-DT) | July 8, 2024 |
| • Certified SolidWorks Professional in Mechanical Design (CSWP) | October 28, 2023 |

Languages

Fluent in English and Arabic (spoken and written)